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First Name(s):															

Department of Mathematics and Applied Mathematics
MAM2000W: 2LA (Linear Algebra)

Test 2	3 May 2023
Time : 1.5 hours	Maximum marks: 52
	Full marks: 50

- This question paper consists of 11 pages (including this one).
- Answer all questions in the spaces provided on this question paper. Use the backs of pages when you run out of space.
- Use your *UCT Examination Answer Book* for rough work. We won't take this work in! Only the answers that you write on this question paper will be marked.
- Basic scientific calculators are allowed.
- Be careful to provide answers that we can read and make sense of at all times. We will pay attention to your presentation as well as the content. Work that is poorly presented will be penalized. If we can't read your handwriting, we will mark your solution incorrect.

DO NOT WRITE BELOW THIS LINE

Q1	Q2	Q3	Q4	Q5	Q6	Q7	Q8	Q9	TOTAL
8	5	6	2	5	8	8	5	5	52

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Question 1. [8 points] Let $A = \begin{pmatrix} 1 & 2 & 3 & 0 \\ 3 & 0 & -1 & 0 \\ 0 & 6 & 10 & 0 \\ 10 & 2 & 0 & 1 \\ 4 & 2 & 2 & 1 \end{pmatrix}$ and $U = \begin{pmatrix} 1 & 2 & 3 & 0 \\ 0 & 6 & 10 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{pmatrix}$.

U is a row-echelon form of A . (You need not check this.)

- (a) Write down a basis for $RS(A)$. (No reason required.)

For instance $\{(1, 2, 3, 0), (0, 6, 10, 0), (0, 0, 0, 1)\}$

- (b) Write down a basis for $CS(A)$. (No reason required.)

For instance $\left\{ \begin{pmatrix} 1 \\ 3 \\ 0 \\ 10 \\ 4 \end{pmatrix}, \begin{pmatrix} 2 \\ 0 \\ 6 \\ 2 \\ 2 \end{pmatrix}, \begin{pmatrix} 0 \\ 0 \\ 0 \\ 1 \\ 1 \end{pmatrix} \right\}$

- (c) Find the nullity and rank of A .

Nullity(A) = 1, Rank(A) = 3

- (d) Find the nullity of A^T .

Nullity(A^T) = $5 - 3 = 2$

- (e) Can every vector $\mathbf{w} \in \mathbb{R}^5$ be expressed in the form $\mathbf{w} = A\mathbf{x}$ for this matrix A and some vector $\mathbf{x}$? Justify your answer.

No.

$\dim(\mathbb{R}^5) = 5$, but $\dim(CS(A)) < 5$.

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(f) Does $A\mathbf{x} = A\mathbf{y}$ (for this matrix A) imply that $\mathbf{x} = \mathbf{y}$? Justify your answer.

No.

A possible justification:

We have $\text{Nullity}(A) = 1$. Hence $NS(A) \neq \{\mathbf{0}\}$.

It follows that the equation

$$A\mathbf{x} = \mathbf{0}$$

has more than one solution.

Hence there are distinct $\mathbf{x}$ and $\mathbf{y}$ such that $A\mathbf{x} = A\mathbf{y}$.

Question 2. [5 points] Let $M_{2 \times 2}(\mathbb{R})$ be the vector space of all 2×2 real matrices, with the usual addition and scalar multiplication for matrices, and let

$$S = \left\{ \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}, \begin{pmatrix} 0 & 1 \\ 1 & 2 \end{pmatrix}, \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}, \begin{pmatrix} 2 & 0 \\ 0 & 0 \end{pmatrix} \right\}.$$

(a) Is S a basis for $M_{2 \times 2}$? Justify your answer.

S is not a basis for $M_{2 \times 2}(\mathbb{R})$.

For instance, we can see that $\begin{pmatrix} 0 & 1 \\ 2 & 0 \end{pmatrix} \notin \text{span}(S)$.

Or we can show that S is not linearly independent.

(b) Find a basis for $\text{span}(S)$.

For instance $\left\{ \begin{pmatrix} 0 & 1 \\ 1 & 2 \end{pmatrix}, \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}, \begin{pmatrix} 2 & 0 \\ 0 & 0 \end{pmatrix} \right\}$

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- (c) An incomplete description of $\text{span}(S)$ is given below; there should be an equation in the place of the dots.

$$\text{span}(S) = \left\{ \begin{pmatrix} a & b \\ c & d \end{pmatrix} \in M_{2 \times 2}(\mathbb{R}) : \dots \right\}$$

Write down this equation.

$$b = c$$

Question 3. [6 points] Let

$$A = \begin{pmatrix} 1 & 0 & -3 \\ 3 & -2 & -3 \\ 0 & 0 & -2 \end{pmatrix}.$$

- (a) Find $\det(A)$.

We may expand along the last row. We obtain

$$\det(A) = (-2) \begin{vmatrix} 1 & 0 \\ 3 & -2 \end{vmatrix} = (-2) \cdot (-2) = 4$$

- (b) Find all the eigenvalues of A .

$$\det(A - \lambda I_3) = 0$$

$$\begin{vmatrix} 1-\lambda & 0 & -3 \\ 3 & -2-\lambda & -3 \\ 0 & 0 & -2-\lambda \end{vmatrix} = 0$$

We expand along the last row and obtain

$$(-2-\lambda) \begin{vmatrix} 1-\lambda & 0 \\ 3 & -2-\lambda \end{vmatrix} = 0$$

$(-2-\lambda)(1-\lambda)(-2-\lambda) = 0$. Hence the eigenvalues are -2 and 1. (-2 is a repeated eigenvalue.)

- (c) Find all the eigenvectors corresponding to the largest eigenvalue of A .

The largest eigenvalue is $\lambda = 1$.

We have

$$A - \lambda I_3 = \begin{pmatrix} 1 - \lambda & 0 & -3 \\ 3 & -2 - \lambda & -3 \\ 0 & 0 & -2 - \lambda \end{pmatrix} = \begin{pmatrix} 0 & 0 & -3 \\ 3 & -3 & -3 \\ 0 & 0 & -3 \end{pmatrix}$$

We solve the equation

$$(A - \lambda I_3)\mathbf{x} = \mathbf{0}$$

$$\begin{pmatrix} 0 & 0 & -3 \\ 3 & -3 & -3 \\ 0 & 0 & -3 \end{pmatrix} \begin{pmatrix} x_1 \\ x_2 \\ x_3 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \\ 0 \end{pmatrix}$$

It follows that

$$\mathbf{x} = s \begin{pmatrix} 1 \\ 1 \\ 0 \end{pmatrix}, \quad s \neq 0.$$

Question 4. [2 points] Let B be a skew-symmetric $n \times n$ matrix, where n is odd. Show that $\det(B) = 0$.

Since B is skew symmetric, it holds that $B = -B^T$.

Hence

$$\det(B) = \det(-B^T). \quad (*)$$

Because n is odd, we have $\det(-B^T) = -\det(B^T) = -\det(B)$.

Therefore, Eq. (*) implies

$$\det(B) = -\det(B)$$

$$2\det(B) = 0$$

$$\det(B) = 0.$$

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Question 5. [5 points] Let

$$p_1(x) = x + 1, \quad p_2(x) = x^2, \quad p_3(x) = x^3 + x$$

$$p_4(x) = x^3 + x^2 - 1, \quad p_5(x) = x^3 - 1, \quad p_6(x) = x^3 - x$$

and $A = \{p_1(x), p_2(x), p_3(x)\}, \quad B = \{p_1(x), p_3(x), p_5(x)\}.$

From the list choose, if possible, a polynomial $p_i(x)$ ($1 \leq i \leq 6$) such that:

(a) $A \cup \{p_i(x)\}$ is a basis for P_3 .

p_6

(the only possible solution)

(b) $B \cup \{p_i(x)\}$ is a basis for P_3 .

Impossible

(since B is linearly dependent)

(c) $A \cup \{p_i(x)\}$ is linearly dependent and $B \setminus \{p_i(x)\}$ is a basis for $\text{span}(B)$.

p_5

(the only solution)

Write **only one of** $p_1(x), p_2(x), p_3(x), p_4(x), p_5(x), p_6(x)$ or “Impossible” in the appropriate block. You may use a polynomial more than once, or not at all.

Question 6. [8 points] Read the following proof and then answer the questions following it:

Proof: Let V be an n -dimensional vector space and $S = \{\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_n\}$ a subset of V such that $\text{span}(S) = V$. Suppose S is linearly dependent. (1)

Then there is an i such that $\mathbf{v}_i$ can be written as a linear combination of the vectors in $S \setminus \{\mathbf{v}_i\}$ (2)

It follows that $V = \text{span}(S \setminus \{\mathbf{v}_i\})$ (3)

But this gives a contradiction. (4)

(a) State the result proved here.

If V is an n -dimensional vector space and S a set in V with exactly n vectors such that $\text{span}(S) = V$, then S is linearly independent (and hence is a basis for V).

(b) Explain how the statement at (3) follows from (2).

Suppose $\exists \mu_1, \dots, \mu_{i-1}, \mu_{i+1}, \dots, \mu_n$ such that

$$\mathbf{v}_i = \mu_1 \mathbf{v}_1 + \mu_2 \mathbf{v}_2 + \dots + \mu_{i-1} \mathbf{v}_{i-1} + \mu_{i+1} \mathbf{v}_{i+1} + \dots + \mu_n \mathbf{v}_n. \quad (*)$$

Let $\mathbf{u}$ be any vector in V . Then $\exists \lambda_1, \lambda_2, \dots, \lambda_n$ such that

$$\mathbf{u} = \lambda_1 \mathbf{v}_1 + \dots + \lambda_n \mathbf{v}_n.$$

We substitute $\mathbf{v}_i$ using Eq. (*) and obtain

$$\mathbf{u} = \lambda_1 \mathbf{v}_1 + \dots + \lambda_{i-1} \mathbf{v}_{i-1} + \lambda_i (\mu_1 \mathbf{v}_1 + \mu_2 \mathbf{v}_2 + \dots + \mu_{i-1} \mathbf{v}_{i-1} + \mu_{i+1} \mathbf{v}_{i+1} + \dots + \mu_n \mathbf{v}_n) + \lambda_{i+1} \mathbf{v}_{i+1} + \dots + \lambda_n \mathbf{v}_n.$$

It follows that $V = \text{span}(S \setminus \{\mathbf{v}_i\})$.

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(c) What is the contradiction referred to at (4)?

If $V = \text{span}(S \setminus \{v_i\})$ then V equals the span of $n-1$ vectors, which is a contradiction to the fact that V is n -dimensional.

(d) Suppose R is a subset of V such that $\text{span}(R) = V$, but that R is not a basis of V . For each of the following statements, either prove it, or disprove it by giving a counter example:

(i) R has more than n elements.

This is true.

1.) Suppose that R has exactly n elements. Then by the result proved above, R is a basis, contradiction.

2.) Suppose that R has less than n elements. Then $\text{span}(R) \neq V$, contradiction.

Hence R must have more than n elements.

(ii) Every subset of R with n elements is a basis for V .

This is false. Counter example: Let $V = \mathbb{R}^3$. Then $\dim(V) = 3$.

Take

$$R = \left\{ \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix}, \begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix}, \begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix}, \begin{pmatrix} 2 \\ 0 \\ 0 \end{pmatrix} \right\}.$$

We have $\text{span}(R) = V$ and R is not a basis for V .

We can see that the following 3-element subset of R is not a basis for V :

$$\left\{ \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix}, \begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix}, \begin{pmatrix} 2 \\ 0 \\ 0 \end{pmatrix} \right\}$$

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Question 7. [8 points] Let the function $T : P_2 \rightarrow P_3$ be defined by $T(p(x)) = xp(x) - x^2p'(x)$, where as usual p' denotes the derivative of p .

(a) Find $T(p(x))$ when $p(x) = ax^2 + bx + c$.

We have $p(x) = ax^2 + bx + c$ and $p'(x) = 2ax + b$.

It follows that

$$\begin{aligned} T(p(x)) &= x(ax^2 + bx + c) - x^2(2ax + b) \\ &= -ax^3 + cx. \end{aligned}$$

(b) Show that T is a linear transformation.

T preserves addition: Let $a_1x^2 + b_1x + c_1$ and $a_2x^2 + b_2x + c_2$ be any vectors in P_2 .

Then we have

$$\begin{aligned} T((a_1x^2 + b_1x + c_1) + (a_2x^2 + b_2x + c_2)) &= T((a_1 + a_2)x^2 + (b_1 + b_2)x + (c_1 + c_2)) \\ &= -(a_1 + a_2)x^3 + (c_1 + c_2)x \end{aligned}$$

← equal

and

$$T(a_1x^2 + b_1x + c_1) + T(a_2x^2 + b_2x + c_2) = -a_1x^3 + c_1x + (-a_2x^3 + c_2x)$$

← equal

T preserves scalar multiplication: Let $ax^2 + bx + c$ be any vector in P_2 and λ any scalar.

Then we have

$$T(\lambda(ax^2 + bx + c)) = T(\lambda ax^2 + \lambda bx + \lambda c) = -(\lambda a)x^3 + (\lambda c)x$$

← equal

and

$$\lambda \cdot T(ax^2 + bx + c) = \lambda(-ax^3 + cx)$$

← equal

(c) Find $\ker(T)$. Show your working.

We need to find a, b, c such that

$$T(ax^2 + bx + c) = -ax^3 + cx = 0$$

Hence

$$a = c = 0.$$

Therefore,

$$\ker(T) = \{bx \mid b \in \mathbb{R}\}.$$

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(d) Is T one-to-one? Give a reason for your answer.

T is not one-to-one because $\ker(T) \neq \{0\}$.

(e) Give, if possible, an example of an element of P_3 which is not in the range of T .

For instance the polynomial x^2 .

Question 8. [5 points] Let $A = \begin{pmatrix} 0 & 1 \\ 2 & 3 \end{pmatrix}$ and V be the vector space of all 2×2 matrices.

Define

$$S = \{B \in V : BA = AB\}.$$

Answer the following questions, giving reasons for all your answers.

(a) Is S closed under scalar multiplication?

S is closed under scalar multiplication:

Let B be any element in S .

Then $BA = AB$.

It follows that for any scalar λ ,

$$(\lambda B)A = \lambda(BA) = \lambda(AB) = A(\lambda B).$$

Hence $\lambda B \in S$.

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(b) Is S closed under addition?

S is closed under addition:

Let B_1 and B_2 be any elements in S .

Then $B_1A = AB_1$ and $B_2A = AB_2$.

It follows that

$$(B_1 + B_2)A = B_1A + B_2A = AB_1 + AB_2 = A(B_1 + B_2).$$

Hence

$$B_1 + B_2 \in S.$$

(c) Is S a subspace of V ?

1.) S is non-empty because it contains (for instance) the matrix I_2 .

2.) S is closed under scalar multiplication.

3.) S is closed under addition.

It follows that S is a subspace of V .

Question 9. [5 points]

(a) Find a matrix A such that $CS(A) = \text{span} \left\{ \begin{pmatrix} 1 \\ 0 \\ 1 \end{pmatrix}, \begin{pmatrix} 1 \\ 2 \\ 0 \end{pmatrix} \right\}$ and $RS(A) = \text{span}\{(1, 0, 1), (1, 2, 0)\}$.

We can for instance take the matrix

$$A = \begin{pmatrix} 1 & 1 \\ 0 & 2 \\ 1 & 0 \end{pmatrix} \begin{pmatrix} 1 & 0 & 1 \\ 1 & 2 & 0 \end{pmatrix} = \begin{pmatrix} 2 & 2 & 1 \\ 2 & 4 & 0 \\ 1 & 0 & 1 \end{pmatrix}.$$

There are many other matrices with these properties.

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(b) Why is it not possible to find a matrix A such that $NS(A) = \text{span} \left\{ \begin{pmatrix} 1 \\ 0 \\ 1 \end{pmatrix}, \begin{pmatrix} 1 \\ 2 \\ 0 \end{pmatrix} \right\}$ and $RS(A) = \text{span}\{(1, 0, 1), (1, 2, 0)\}$?

Suppose A is a matrix with these properties.

Because $RS(A) = \text{span}\{(1, 0, 1), (1, 2, 0)\}$, A must have 3 columns.

On the other hand, the number of columns in A equals

$$\text{Rank}(A) + \text{Nullity}(A) = 2 + 2 = 4.$$

This is a contradiction.

Therefore, such a matrix A does not exist.